Lord of the Wings [2026]

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Quest 5: The Cartograffity

‹ Quest 4 Quest 6 ›

Part I

> **Story section**
>
> Once the tournament results had finally been calculated, the elves officially declared Legolink Pointereye their greatest warrior. As promised, he joined our expedition. Before we set off, the Grand Council handed us several maps to guide us safely to Nulldor.
>
> With an elven warrior leading the way, we expected a smooth journey. During our very first stop, however, Frodoff picked up one of the maps. He turned it one way, then the other, held it upside down, frowned, and started the whole process again.
>
> Gimlock watched this strange performance for a moment before suggesting that the elves had probably made another perfectly sensible invention. Perhaps the missing details appeared only to people who already knew where they were going.
>
> It turned out that neither explanation was correct. The route was perfectly visible, and the destination had been marked just as clearly. The problem was that nothing else was on the map. There were only lighter and darker patches, like the static on an old crystal ball that had lost its signal.
>
> Gimlock stared at the map for a while before handing it back. "At least it's consistently elven." Since we would all need to use these maps on the journey to Nulldor, we agreed that everyone should learn how to read them. Legolink offered to explain everything from the beginning.

Legolink explained that the lighter and darker patches were not random at all. To make the maps easier to read, we could represent every lighter cell as `0` and every darker cell as `1` (your notes). Every map had an odd number of rows and an odd number of columns. This meant that it always had exactly one central cell.

To reconstruct the land, **begin by drawing a single land cell,** represented by `#`, **in the exact centre of the map.** From that point, new pieces of land are gradually added around the land that has already been discovered. **Consider every cell that touches the current land** horizontally, vertically or diagonally.

A cell becomes land if at least `5` of its eight neighbouring cells are either already marked as land `#`, or dark cells `1` on the original map. Whenever a cell meets this condition, replace it with `#`. It becomes part of the known land and may allow more cells around it to be attached. Continue until no more land cells can be added.

Example based on the following notes:

```
010100100100001101000
111101001000001110000
101111011000101010011
001000101010000100000
000110100110011001100
000001110111111000000
101100010101011110000
001101100000001010011
001001001111010001001
100011011110011010101
001100101110010010000
```

The first step is to mark the central cell of the map as land `#`. The diagrams below highlight all cells adjacent to the known land, including those touching only at a corner, as these are the only cells that can be added to the land next.

```
010100100100001101000
111101001000001110000
101111011000101010011
001000101010000100000
000110100110011001100
0000011101#1111000000
101100010101011110000
001101100000001010011
001001001111010001001
100011011110011010101
001100101110010010000
```

After repeatedly attaching every possible cell, the reconstructed land looks like this:

```
010100100100001101000   010100100100001101000   010100100100001101000   010100100100001101000
111101001000001110000   111101001000001110000   111101001000001110000   111101001000001110000
101111011000101010011   101111011000101010011   101111011#00101010011   10111101##00101010011
001000101010000100000   001000101#10000100000   001000101#1#0#0100000   0010001########100000
000110100###011001100   000110100####11001100   0001101#######1001100   000110#########001100
0000011101#1111000000   00000111#####11000000   00000111######1000000   000001#########000000
1011000101#1011110000   1011000101#1011110000   10110001######1110000   101100########1110000
001101100000001010011   001101100##0001010011   00110110####001010011   0011011#######1010011
001001001111010001001   001001001111010001001   00100100###1010001001   0010010####1010001001
100011011110011010101   100011011110011010101   100011011110011010101   10001101###0011010101
001100101110010010000   001100101110010010000   001100101110010010000   001100101110010010000

010100100100001101000   010100100100001101000   010100100100001101000   01#100100100001101000
111101001000001110000   111101##10000####0000   111######0000####0000   11#######0000####0000
101111####001###10011   1011######001###10011   101#######00####10011   10########00#####0011
00100##########100000   0010############00000   001##############0000   001##############0000
00011##########001100   00011###########01100   0001############01100   000##############1100
000001##########00000   0000############00000   000##############0000   000##############0000
101100##########10000   10110###########10000   1011############10000   101##############0000
001101##########10011   001101##########10011   0011#############0011   001##############0011
001001#########001001   00100###########01001   0010#############1001   001##############1001
100011#####0011010101   10001#########1010101   10001###########10101   100#1#############101
001100101#10010010000   0011001#1#10010010000   0011001###10010010000   0011001###10010010000

01#100100100001101000   01#100100100001101000   ..#..................
1########0000####0000   1########0000####0000   .########....####....
1#########00#####0011   1#########00#####0011   .#########..#####....
001###############000   00################000   ..################...
000##############1100   000###############100   ...###############...
000###############000   000###############000   ...###############...
101##############0000   101##############0000   ...##############....
00###############0011   00###############0011   ..###############....
001###############001   00################001   ..################...
10################101   10################101   ..################...
0011001###10010010000   0011001###10010010000   .......###...........
```

The final diagram shows only land cells as `#` and water cells as `.` for better readability.
The reconstructed land contains `137` cells.

How many cells on the map are occupied by land?

Your notes for this part: Copy Open Download

Part I solved with answer: 1280
Check your progress

Part II

Gimlock was still not impressed. He pointed out that such a map could not even tell the difference between solid land and a sandy beach. Legolink calmly picked up another map (your notes) and assured us that it could.

First, reconstruct the land exactly as in the previous part. Once the entire land surface has been found, the sandy coastline can be determined.

Every land cell that is **adjacent to at least one water cell**, including diagonal neighbours, immediately becomes sand `=`. Always treat the map as if it were surrounded by water on all sides, so every cell on the edge of the map is considered adjacent to water. The remaining sand is then found in much the same way as the land was in the previous part. **A land cell becomes sand** if at least `5` of its eight neighbouring cells are already marked as sand `=`. Continue until no more cells can become sand.

Example based on the following notes:

```
010100100100001101000
111101001000001110000
101111011000101010011
001000101010000100000
000110100110011001100
000001110111111000000
101100010101011110000
001101100000001010011
001001001111010001001
100011011110011010101
001100101110010010000
```

This is exactly the same example as in the previous part, so we already know what the reconstructed land looks like.

```
..#..................
.########....####....
.#########..#####....
..################...
...###############...
...###############...
...##############....
..###############....
..################...
..################...
.......###...........
```

First, every land cell adjacent to water becomes sand `=`. This includes diagonal neighbours. The map is also treated as if it were surrounded by water on all sides, so every land cell on its edge becomes sand as well.

```
..=..................
.========....====....
.==#####==..==##=....
..==#####====###==...
...=#############=...
...=############==...
...=############=....
..==############=....
..=#############==...
..======#=========...
.......===...........
```

The remaining sand is found in much the same way as the land in the previous part. During each step, every land cell with at least `5` neighbouring sand cells also becomes sand.

```
..=..................   ..=..................   ..=..................
.========....====....   .========....====....   .========....====....
.===####==..==#==....   .====###==..=====....   .====###==..=====....
..==#####====###==...   ..==#####====###==...   ..==#####====##===...
...=############==...   ...=############==...   ...=############==...
...=############==...   ...=############==...   ...=############==...
...=############=....   ...=############=....   ...=############=....
..==############=....   ..==############=....   ..==############=....
..==###########===...   ..===##########===...   ..===##########===...
..================...   ..================...   ..================...
.......===...........   .......===...........   .......===...........
```

After the third growth step, no more land cells have at least five sand neighbours, so the process stops.
The reconstructed beach contains `69` sand cells.

How many cells on the map are occupied by sand?

Your notes for this part: Copy Open Download

Part II solved with answer: 1383
Check your progress

Part III

Gimlock stared at the latest map in complete disbelief. "Don't tell me these maps can even show trees. What next, potato stamps?" Legolink quietly opened a small wooden chest filled with... potato stamps. Gimlock blinked. Without a word, Legolink picked up a potato stamp carved into the shape of a tree, then took out another map (your notes) to demonstrate the procedure.

First, reconstruct the land and sand exactly as before. Once both have been determined, the remaining land can be decorated with trees. **A tree can be added at X whenever the following 5 by 5 pattern fits:**

```
.+++.
+++++
++X++
+++++
.+++.
```

For every land cell, treat that cell as `X` and examine the area around it:

Dots `.` represent cells that are irrelevant. Every `+` marks a cell that must contain either land `#` or a tree `@`. If all cells marked with `+` meet this condition, the land cell at `X` becomes a tree `@`. Check every land cell in this way.

Once all possible trees have been added with the stamp, **the forest grows for exactly two more steps**. During each step, a land cell becomes a tree if:

- at least `4` of its eight neighbouring cells are already trees, and
- none of its eight neighbouring cells is sand.

Each of these two steps is applied simultaneously across the entire map. This means that trees added during a step may only affect the following step.

Example based on the following notes:

```
010100100100001101000
111101001000001110000
101111011000101010011
001000101010000100000
000110100110011001100
000001110111111000000
101100010101011110000
001101100000001010011
001001001111010001001
100011011110011010101
001100101110010010000
```

This is the same map as in the previous parts, so the land and sand have already been determined. Legolink now checks the remaining land cells from top to bottom and from left to right. Whenever all cells marked with `+` contain land or trees, the cell at `X` becomes a tree `@`. Trees added earlier may therefore make later stamp positions valid.

The maps below show every successful use of the stamp. The two forest-growth steps are shown separately afterwards.

```
..=..................   ..=..................   ..=..................   ..=..................
.========....====....   .========....====....   .========....====....   .========....====....
.====###==..=====....   .====###==..=====....   .====###==..=====....   .====###==..=====....
..==#####====##===...   ..==#####====##===...   ..==#####====##===...   ..==#####====##===...
...=##@#########==...   ...=##@#########==...   ...=##@#########==...   ...=##@#########==...
...=############==...   ...=##@#########==...   ...=##@@########==...   ...=##@@########==...
...=############=....   ...=############=....   ...=############=....   ...=##@#########=....
..==############=....   ..==############=....   ..==############=....   ..==############=....
..===##########===...   ..===##########===...   ..===##########===...   ..===##########===...
..================...   ..================...   ..================...   ..================...
.......===...........   .......===...........   .......===...........   .......===...........

..=..................   ..=..................   ..=..................   ..=..................
.========....====....   .========....====....   .========....====....   .========....====....
.====###==..=====....   .====###==..=====....   .====###==..=====....   .====###==..=====....
..==#####====##===...   ..==#####====##===...   ..==#####====##===...   ..==#####====##===...
...=##@#########==...   ...=##@#########==...   ...=##@#########==...   ...=##@#########==...
...=##@@########==...   ...=##@@########==...   ...=##@@########==...   ...=##@@########==...
...=##@@########=....   ...=##@@@#######=....   ...=##@@@@######=....   ...=##@@@@@#####=....
..==############=....   ..==############=....   ..==############=....   ..==############=....
..===##########===...   ..===##########===...   ..===##########===...   ..===##########===...
..================...   ..================...   ..================...   ..================...
.......===...........   .......===...........   .......===...........   .......===...........

..=..................   ..=..................   ..=..................
.========....====....   .========....====....   .========....====....
.====###==..=====....   .====###==..=====....   .====###==..=====....
..==#####====##===...   ..==#####====##===...   ..==#####====##===...
...=##@#########==...   ...=##@#########==...   ...=##@#########==...
...=##@@########==...   ...=##@@########==...   ...=##@@########==...
...=##@@@@@@####=....   ...=##@@@@@@@###=....   ...=##@@@@@@@@##=....
..==############=....   ..==############=....   ..==############=....
..===##########===...   ..===##########===...   ..===##########===...
..================...   ..================...   ..================...
.......===...........   .......===...........   .......===...........
```

During the first growth step, `1` additional land cell becomes tree:

```
..=..................
.========....====....
.====###==..=====....
..==#####====##===...
...=##@#########==...
...=##@@@#######==...
...=##@@@@@@@@##=....
..==############=....
..===##########===...
..================...
.......===...........
```

During the second growth step, `2` additional land cells become trees:

```
..=..................
.========....====....
.====###==..=====....
..==#####====##===...
...=##@@########==...
...=##@@@@######==...
...=##@@@@@@@@##=....
..==############=....
..===##########===...
..================...
.......===...........
```

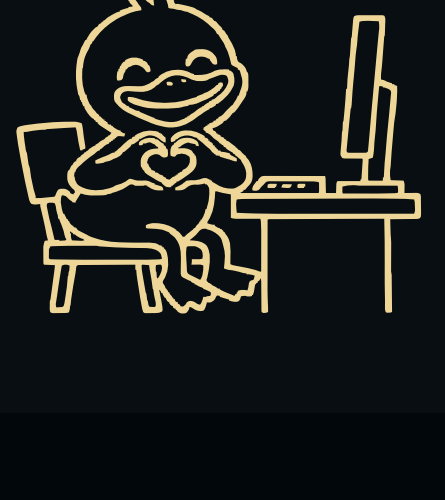
The final map contains `14` tree cells.

How many cells on the map are occupied by trees?

Your notes for this part: Copy Open Download

Part III solved with answer: 9865
Check your progress

Puzzle solved! Don't stop now!
Post your solution, compare ideas, and help others grow on Reddit or join us for a chat on Discord



Bonus for Supporters!

This feature is available only to Supporters as a small "thank you" for supporting this event. Click the link below to **get all 100 versions of the input notes prepared for this Quest**, along with the answers! Please don't host it anywhere; especially please don't add it to your solutions repo. Thank you for being awesome! ❤

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